

$$\frac{713.4 \times 1.044}{9.3\% - 9\%} =$$

$$\frac{713.4 \times 1.044}{(1.093)^{10}} \div 565.2 = 11.05$$

$$54.29 - 11.05 = 43.24$$

→ First Mistake

$g = 3\%$
 $NI = 0$ } → Does not make sense.

Terminal Year

This relationship must hold

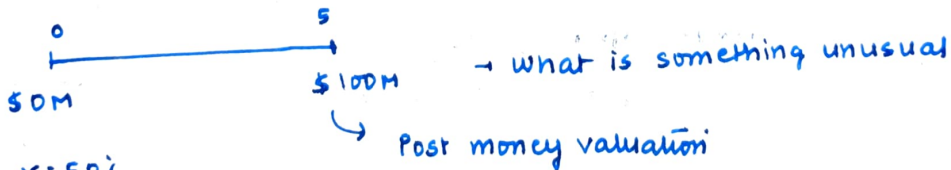
$$g = \text{ADIC} \times \text{Inv. Rate}$$

12th August 2026

Lecture - 16

The venture capital problem set

(a.) what share of the company will a venture capitalist required rate of return is 50% per annum.



$r = 50\%$

* what is different?

So far growth was being funded internally.

Illiquidity

Premium

$K_e \rightarrow$ Function of systematic risk.

* Venture Capital and Credit Rating highest rated.

Founders 1M shares

↳ How many shares to VC, to raise that 5 million?

$$5 \times 1.5^5 = 37.97 \text{ [38\%]}$$

Stake: 37.97 from the business.

$$\frac{100}{1.5^5} = 13.17 \quad 13.17 - 5 = 8.17$$

Postmoney

8.17 value of 1 million shares

Value of 1 share = $\frac{8.17}{1}$

Alternatively

$$5M \text{ for } 37.97\% \text{ stake} = 13.17M$$

→ 62% shares is basically 1 Million [left with the founder]

$$\frac{1M}{0.6203} = 1.612M$$

0.612 Million shares for 5 million

$$\frac{5}{0.612} = 8.17$$

→ In the startup valuation, all the money goes to the company.

⇒ Assume investment is in standard preferred stock with no dividends and a conversion ratio to common 1:1.

→ Preferred stock to have
safeguard against liquidity

Similar to Equity shares.

$$c.) \quad 12 \times 1.5^5 = 91.25$$

$$100 - 91.25 = 8.75$$

$$\frac{1}{0.8875} = 11.26M$$

e.) Management gets 15% share of the company.

$$5M \times 1.5^5 = 37.97M$$

38% stake to VC $\rightarrow$ will he be happy?
(No)

$$38\% \rightarrow 38\% \text{ of } 85$$

15% belong to management

$$\underline{\underline{44.7\%}}$$

$$38\% = \frac{x \times 85}{100}$$

$\rightarrow$ 44.68% should be given to the VC so that he becomes happy 😊

$\rightarrow$ Another way to get the 44.68%.

Everything remaining same

$$\frac{37.97\%}{1 - \text{dilution}\%} = \frac{37.97\%}{1 - 15\%}$$

1 Million share is giving the founder 1 - 44.68% control.

$$\frac{1}{1 - 44.68\%} = 1.81M \text{ [Total No. of shares]}$$

$$\rightarrow SP = \frac{\$5M}{0.81M} = \$6.19 / \text{share} \text{ [For VC]}$$

VC saying I want 38% in the business, post the dilution

$\rightarrow$ Share price 5 years from now?

$$\frac{1.81M}{0.85} = 2.12 \text{ [No. of shares]}$$

$$\frac{100}{2.12} = 46.94 \text{ per share value.}$$

→ Take care of dilution before next round of valuation

a
Part 2.1

$$5 \text{ Million} \rightarrow \frac{37.97}{(5)} = 32.97$$

$$5M + x\% \text{ of } 95 = 37.97$$

$$\frac{37.97}{95} = 34.74\%$$

What would be the stock price today?

$$\frac{1}{1 - 34.74\%} =$$

~~5 Million~~ shares. Overall no. of shares

$$\frac{5}{\text{Overall no. of shares} - 1} = \$9.39/\text{Share}$$

Question 3.0



VC₁ = 5M
r = 50%
n = ?
p = ?

VC₂ = 3M
r = 30%
n = ?
p = ?

VC₃ = 5M
mgmt = 15%
n = ?
p = ?

$$(1.5)^5 \times 5 \rightarrow 38\%$$

$$(1.3)^3 \times \text{?} \rightarrow 6.591\%$$

$$\frac{100}{85}$$

VC₁ → \$38M
VC₂ → \$6.59M
mgmt → \$15M

Founder → 40.41M

$$\text{Stake VC}_1 = \frac{38}{38 + 40.41}$$

$$\text{Stake VC}_2 = \frac{6.59}{85} = 7.75\%$$

only VC and founder are there.

without any dilution = 33.97%.

↓

After 1 dilution

44.68%.

↓

$$\frac{44.68\%}{1 - 7.75\%} = \underline{\underline{58\%}}$$

$$\frac{6.59\%}{1 - 15\%} = 7.75\%$$

Year #0

n = 1.938M

Founder → 1M

VC1 → 0.938

$$P = \frac{1}{\underline{\quad}} = \underline{1.93\%}$$

~~1.58%~~
(1.58%)

$$\frac{5M}{0.938} =$$

Year #2

n =

Founder

VC1

VC2

$$\frac{1.938}{9.2725\%} = 2.10$$

$$\frac{34}{6.162M} = \$18.40$$

Year #5

n =

Founder

VC1

VC2

mgmt

$$\frac{2.10}{0.85} = 2.474$$

14th August 2026

Lecture-17

$$EVA = 0$$

$g = \text{Retention Ratio} \times ROE \rightarrow \text{Growth rate in profit}$
 $\downarrow$
 constant
 $\downarrow$
 Growth in Dividends

→ But this equation does not hold in the initial years, holds value during terminal years.

$$IA = \frac{\text{Net Investment}}{NOPAT}$$

$$g = ROIC \times \text{Investment rate}$$

→ This equation must be followed or terminal value would be incorrect.

$$EV = OIC + PV(EVA)$$

Value using EVA method

$$ROIC (k) = 20\%$$

$NOPAT = \text{ROI} \times OIC$
 $\downarrow$ $\downarrow$
 o/p of financial model output of the project not the input

FCF method, $\frac{7}{1.13} + \frac{10.6}{1.13^2} + \frac{15}{1.13^2} + \frac{15(1.05)}{13\% - 5\%}$
 $= 161.34$

EVA method, $100 + 7 + 7.91$

$$EVA_t = DIC_{t-1} (ROI_{t-1} - WACC_t)$$

$$FCF = NOPAT_t \cdot NI_t$$

$$g = ROIC \times IR$$

↳ Must hold in terminal year.

But changing the g directly is incorrect, as it is an input to the model.

→ You actually invested more than ~~is~~ required to grow at g .

→ Restate year 4 numbers

↳ Forecast for year 4. $(t+2)$ years.

EVA method estimate for $t+1$ years.

→ How terminal value and terminal growth rate are related.

Key Difference: Public vs Private valuation

↑ 4 points of difference ↓

• Financial data, not that readily available for private company.

• Market Price

• Ownership Structure

• Liquidity.

• Corporate Governance tells problems which are going to come.

- Market value not available $\rightarrow \beta$ cannot be calculated
- Debt ratios for going from unlevered to levered
- computing cost of capital
- cash flow estimation issues.
 - Different Accounting standards
 - Personal and Business expenses
 - Salaries from dividends
 - Pass through entities: no entity level tax

Private company

Adjustment in Financial statements

- Profit figures might be overstated.
- Assets: Personal and Business Assets.
- Private company financials often need adjustments for owner compensation, personal expenses and non-recurring items.

→ Salary for ghost employees, personal travel expenses, attending seminars, high salary for CEO.

Control Premiums 20-30% premium for the ability to control financial operations

Discount Rate and Cost of Equity

Private company Valuation

Private to
Private
Transactions

Private to
Public
transactions

Private to
IPO

systematic &
unsystematic
risk

Key difference

- Illiquidity should not apply
- Cost of capital different

Private to private transaction

- Key person value → value of person inversely proportional to the stature of the key person in the company.
- Neither buyer or seller is diversified.

The buyer buys the company minus key person.

Valuing Kavita Textiles

- Kavita and Husband taking salaries,
 - Selling and Distribution expenses and admin. Expenses are probably overs
- Use market model.

$$r_{i,t} = \alpha_i + \beta_i \times r_{m,t} + \text{errors}$$

→ Systematic risk.

$$\sigma_i^2 = \beta_i^2 \times \sigma_m^2 + \sigma^2_{\text{errors}}$$

→ Non systematic risk

How risk is divided b/w systematic and non systematic risk.

→ When D/E ratio changes WACC changes.

Equity Value 345.67, This is post money value.

Pre-money value 20.9.25 cr
- 50 Equity

Post Money 500 cr

200 cr

- 50 cr Debt
- 50 cr employees
- 100 cr company

Issue Price = $\frac{400 \text{ cr}}{\text{n of existing shares}}$

Pre-money = 400 cr

19th August 2026

Lecture 18

AI in Business Valuation

Business Analysis and Valuation

How AI can help: what changes and what doesn't

- Valuation theory is unchanged
- DCF, multiples, cost of capital, terminal value

→ What changes

How AI has evolved: From Prompts to Agents

Grok 4.6 → GrokBot (Best)

Breaks a goal into steps, calls tools and data sources, checks its outputs and keeps going without a prompt for

Custom GPT in OpenAI

↳ DCF Valuator

Minimax writes the python code.

Warren AI from Investing.com

Cluster Analysis

* EPS can include one off gains



tokens
p / token

How Anthropic makes money

Project P growth rate and Q growth rate individually.

→ Terminal EBITDA multiple

$$TV = \frac{80 \times (1.05)}{14\% - 5\%} \div 200 = 6\%$$

21st August 2026

Lecture- 19

Darshan Rathod

Practical Insights on Valuation

How interesting valuation can be?

Manufacturing and Trading business

→ Fair value of the two companies

Valuation Approaches and Methods

1. Market Approach
2. Income Approach
3. Cost Approach
 - Summation cost method
 - Replacement cost method
 - Reproduction cost method

→ Data challenges & common practices in valuation

Valuation → when is it required

Equity Research	Corporate Finance	Statutory Requirement	Internal
• Sell Side	• M&A	• Income tax Act	• Internal restructuring
	• Fund raising	• Companies Act	• Intangibles
	• IPO	• SEBI	• Indian As.
		• FEMA	
		• IBC	

2nd AS private company < 250cr * Leases • is very different.
↳ ~~titers~~ More flexibility on a fair value basis.

→ Angel Tax

• Gifting no tax is allowed

↳ More than 50,000 amount gift it is taxed.

9(1)(1) → Retrospective Tax



↳ Tax anything, anytime → Lot of Backlash

Vodafone Tax

(Most Important Factors)

Income Approach

Projections

Discounting Factors in DCF method

Practical problems



Kd + (some premium on it)

* If market does not have the liquidity, then the Equity calculated is not correct.

• CAPM

• opportunity cost

• Participant Expectation

Rf → Treasury Bonds.

β → Comparative Analysis.
[Subjectivity involved]

$$K_e = R_f + \beta (R_f - R_m) + \text{CSR}$$



company specific Risk Premium.

cover / Save Your Ass

Ashwath Damodaran

Cons published only once a year on 5th Jan.

* How do you calculate market risk premium?

• Market is on a high MRP ↑

→ Monthly 10 year difference and then average it out.

(6% - 7%) → Generally used market risk premium.

• How do you calculate CSRP?

* Maximum CSRP of 4%.

Different scenarios

WACC →

weight of Equity

Market value of debt and equity.

• Target debt to capital ratio.

• Industry debt to capital ratio.

FCFF or FCFE?

* More used in industry.

Discount factor

keeps on changing.

* Horizon CFs

* Terminal Value → Gordon growth rate.

↳ Business doing a lot of capex



Not fair to use GM model

• H model.

• multi period cash flow.

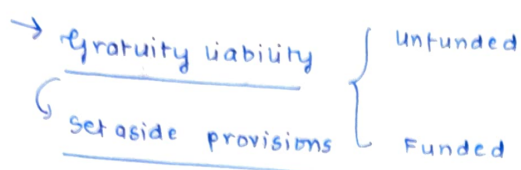
Enterprise value to Equity value.

$$\text{Equity Value} = \text{Enterprise Value} - \text{Debt and debt like items} + \text{surplus assets} - \text{Contingent liability.}$$

Stretch creditors → credit period 60 days

180 days liabilities

* 120 days it becomes debt sitting like item.



Surplus Assets:

Practical problems in Market Approach

- Adjusted EBITDA (most used terminology in Deal making)
- While 100% Equity was bought, 60% was given.
- Information is not available correctly
- * Comparable Transactions.

→ Other way in which we can find out comparable multiples.
Comparable Transaction Multiples.

• Comparable Companies Method: → Discounts and Premiums.

Challenges in Merger-Acquisition and Equity Research

DCF or Multiplier what does the Equity Research do?

Equity Research Valuation: → PVA.

- Discretionary Peers.

PEG ratio of D-mart (4.5) → Justifiable because comparing with Walmart, Target.

* How is M&A valuation different from equity research valuation?

Typical valuation construct in M&A

What are the advantages of using a revenue based multiple?

→ EV / ^{cash proxy} EBITDA multiple of 'x' on sustainable EBITDA on a cash-free, debt-free basis adjusted for normalized working capital and any other non-operating asset.

Normalized Working Capital The sustainable amount of operating working capital required to run business normally, excluding temporary or abnormal fluctuations

